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DEPARTMENT OF COMPUTER ENGINEERING

Laboratory Manual

Class: B.E. Computer

Subject: Advanced Technology Lab-II

Academic Year: 2025-26

Semester: VIII

DEPARTMENT OF COMPUTER ENGINEERING

vision

To emerge as the leading Computer Engineering department for inclusive development of students.

Mission

To provide student-centered conducive environment for preparing knowledgeable, competent and value-added computer engineers.

Program Education Objective:

- **PEO 1. Core Knowledge** -Computer engineering graduates will have the knowledge of basic science and Engineering skills, Humanities, social science, management and conceptual and practical understanding of core computer engineering area with project development.
- **PEO 2. Employment/ Continuing Education** - Computer engineering graduates will have the knowledge of Industry-based technical skills to succeed in entry level engineering position at various industries as well as in academics.
- **PEO 3. Professional Competency** - Computer engineering graduates will have the ability to communicate effectively in English, to accumulate and disseminate the knowledge and to work effectively in a team with a sense of social awareness.

Program outcome

Engineering Graduates will be able to:

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcome:

Computer Engineering Graduates will be able to::

- **Software Systems Development:** Apply the theoretical concepts of computer engineering and practical knowledge in analysis, design and development of software systems.
- **Open-Source Software:** Demonstrate familiarity and practical competence with a broad range of programming languages and open-source platforms
- **Computer Proficiency:** Exhibit proficiency through latest technologies in demonstrating the ability for work efficacy to the industry & society

Course outcomes:

Upon successful completion of lab Course, student will be able to:

1. Break down real world problems / application.
2. Demonstrate Full Stack development.
3. Design Full Stack based applications.
4. Decide tools for Full Stack development.
5. Develop Full Stack based applications.

• CO-PO-PSO Mapping for Advanced Technology Lab -II

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	2	1	2		2	1	2	2		1	1	3	3	2	3
CO2	2		3		2	1	2	2			1	3	3	2	3
CO3	2		3		2	1	2	2			1	3	3	2	3
CO4	2		3		2	1	2	2			1	3	3	2	3
CO5	2	1	3		2	1	2	2			1	3	3	2	3
	2	1	2.8		2	1	2	2		1	1	3	3	2	3

SSBT's College of Engineering & Technology, Bambhori, Jalgaon

Department of Computer Engineering

Academic Year 2021-2022 (Term-II)

Subject: Advanced Technology Lab-II

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Sr. No.	Name of Experiment	Page No	Date of Performance	Date of Completion	Subject Teacher Sign
1	Data Visualization using Python.				
2	Implementation of Django stack				
3	Create a Ruby on Rails application				